

## Solutions (in English)

### Exercise 2 (Multiple testing)

**Statement.** Suppose that we test  $m$  hypotheses, and control the Type I error for each hypothesis at level  $\alpha$ . Assume that all  $m$   $p$ -values are independent, and that all null hypotheses are true.

- (a) Let the random variable  $A_j$  equal 1 if the  $j$ th null hypothesis is rejected, and 0 otherwise. What is the distribution of  $A_j$ ?
- (b) What is the distribution of  $\sum_{j=1}^m A_j$ ?
- (c) What is the standard deviation of the number of Type I errors that we will make?

### Solution

Since all null hypotheses are true and each test is conducted at level  $\alpha$ , for each  $j$  we have

$$\mathbb{P}(A_j = 1) = \alpha, \quad \mathbb{P}(A_j = 0) = 1 - \alpha.$$

- (a) Therefore,

$$A_j \sim \text{Bernoulli}(\alpha).$$

- (b) Because the  $p$ -values are independent, the indicators  $\{A_j\}_{j=1}^m$  are independent Bernoulli( $\alpha$ ) random variables, so their sum has a binomial distribution:

$$\sum_{j=1}^m A_j \sim \text{Binomial}(m, \alpha).$$

- (c) If  $S = \sum_{j=1}^m A_j \sim \text{Binomial}(m, \alpha)$ , then

$$\text{Var}(S) = m\alpha(1 - \alpha),$$

and hence the standard deviation is

$$\text{SD}(S) = \sqrt{m\alpha(1 - \alpha)}.$$

### Exercise 7 (Carseats, ISLP)

**Statement.** This problem makes use of the `Carseats` dataset in the `ISLP` package.

- (a) For each quantitative variable in the dataset besides **Sales**, fit a linear model to predict **Sales** using that quantitative variable. Report the  $p$ -values associated with the coefficient for the variable. That is, for each model of the form  $Y = \beta_0 + \beta_1 X + \varepsilon$ , report the  $p$ -value associated with the null hypothesis  $H_0 : \beta_1 = 0$ . Here,  $Y$  represents **Sales** and  $X$  represents one of the other quantitative variables.
- (b) Suppose we control the Type I error at level  $\alpha = 0.05$  for the  $p$ -values obtained in (a). Which null hypotheses do we reject?
- (c) Now suppose we control the FWER at level 0.05 for the  $p$ -values. Which null hypotheses do we reject?
- (d) Finally, suppose that we control the FDR at level 0.2 for the  $p$ -values. Which null hypotheses do we reject?

## Solution

### (a) Fit simple regressions and collect $p$ -values

The quantitative predictors in **Carseats** (excluding **Sales**) are:

CompPrice, Income, Advertising, Population, Price, Age, Education.

For each predictor  $X$ , fit  $\text{Sales} = \beta_0 + \beta_1 X + \varepsilon$  and extract the  $p$ -value for  $\beta_1$ .

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from ISLP import load_data

Carseats = load_data("Carseats").dropna()

y = Carseats["Sales"].to_numpy()
predictors = ["CompPrice", "Income", "Advertising", "Population", "Price", "Age",
              "Education"]

pvals = []
for xname in predictors:
    X = sm.add_constant(Carseats[xname].to_numpy())
    model = sm.OLS(y, X).fit()
    pvals.append(model.pvalues[1])

out = pd.DataFrame({"variable": predictors, "p_value": pvals}).sort_values(
    ("p_value")
)
print(out)
```

### (b) Per-hypothesis control at $\alpha = 0.05$

Reject  $H_0 : \beta_1 = 0$  for those predictors whose simple-regression  $p$ -value satisfies  $p \leq 0.05$ .

```
alpha = 0.05
rejected_b = out.loc[out["p_value"] <= alpha, "variable"].tolist()
print(rejected_b)
```

### (c) Control FWER at 0.05 (Bonferroni)

With  $m = 7$  tests, Bonferroni uses the threshold  $\alpha/m$ . Reject when  $p \leq 0.05/7$ .

```
m = len(predictors)
thr_bonf = 0.05 / m
rejected_c = out.loc[out["p_value"] <= thr_bonf, "variable"].tolist()
print(thr_bonf)
print(rejected_c)
```

### (d) Control FDR at 0.2 (Benjamini–Hochberg)

Sort the  $p$ -values  $p_{(1)} \leq \dots \leq p_{(m)}$  and find

$$k = \max \left\{ i : p_{(i)} \leq \frac{i}{m} q \right\}, \quad q = 0.2,$$

then reject the  $k$  hypotheses corresponding to the  $k$  smallest  $p$ -values.

```
q = 0.2
p_sorted = out["p_value"].to_numpy()
m = len(p_sorted)
thresholds = (np.arange(1, m+1) / m) * q
k = np.where(p_sorted <= thresholds)[0]
k = (k.max() + 1) if len(k) else 0
rejected_d = out.iloc[:k]["variable"].tolist()
print(rejected_d)
```